Friedrich-Alexander-Universität
Technische Fakultät





AI in Medicine

人工智能在医学中的应用

Prof. Dr.-Ing. habil. Andreas Maier, Pattern Recognition Lab

Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany 30 October 2026 ▪ Shanghai Chest Hospital ▪ 上海市胸科医院

About the Speaker

Prof. Dr.-Ing. Andreas Maier 主讲人简介

Responsibilities & Experience

- Head of the Pattern Recognition Lab, FAU
- Head of the Competence Center Medical Engineering Erlangen
- Head of the Medical AI Hub of Bavaria
- Member, AI Council of the Free State of Bavaria
- Scientific Counselor, Fraunhofer Institute
- AI Council, German Radiation Protection Society
- European Research Council (grantee & reviewer)
- Visiting Professor at Stanford University and MIT

Industry & Entrepreneurship

- Siemens Healthineers (innovation manager / tech lead)
- Start-ups: *Mira Vision* (shareholder, valued at EUR 10 M); *Norizon* (council, early VC phase)
- Holds more than 15 active US and EU patents

Theory first, application always.



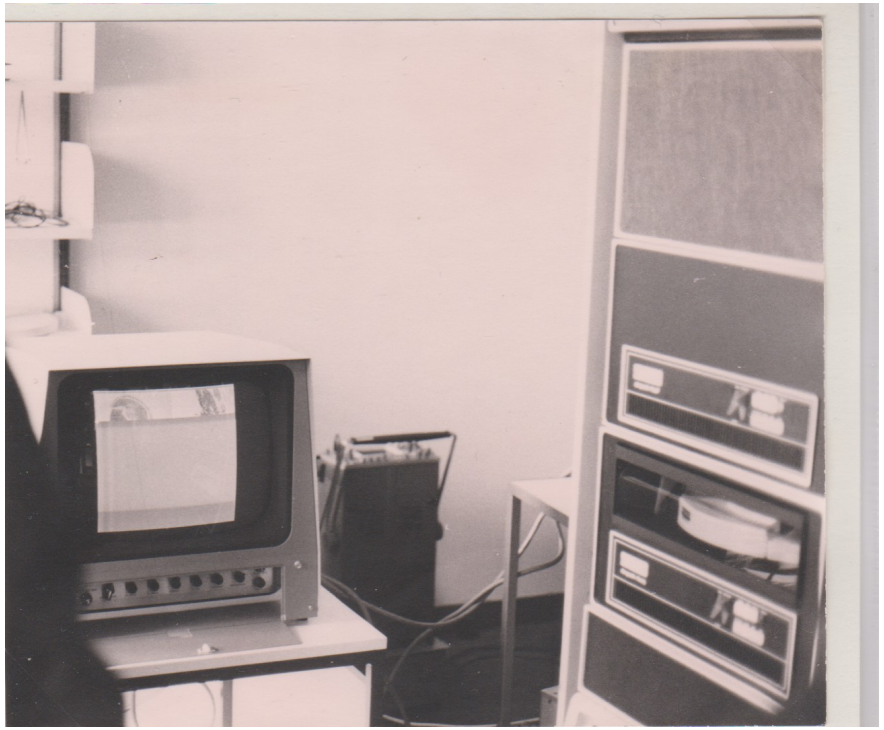
Pattern Recognition Lab

Friedrich-Alexander-Universität Erlangen-Nürnberg 模式识别实验室

Know the Past to Understand the Future

温故而知新

- Founded in 1975, founder: Heinrich Niemann
- Germany's first dedicated Pattern Recognition / Machine Learning Lab
- Medical imaging as early as 1981
- First computer: PDP-11/34 with 248 kB, 2.5 MHz
- Focus: theory & good understanding of hardware



Our Team Today

Pattern Recognition Lab 我们的团队

Secretaries



Alexandra H.



Claudia S.

Head of the Lab



Prof. Dr.-Ing. Andreas Maier

Technician



Sven G.

- ∼60 PhD students & 8 postdocs
- 2 emeritus professors
- ∼3 709 master & bachelor students

- Academic partners in Europe, Asia, North/South America, Australia
- Industry partners: Siemens, Fraunhofer, Adidas, Intel, Audi, IBM, and more

Research Areas

Pattern Recognition Lab 研究方向

Computer Vision

Time Series Intelligence

Magnetic Resonance Imaging

Data Processing for Utility Infrastructure

Learning Algorithms for Medical Big Data Analysis

Speech Processing & Language Understanding

Inverse Problems & Applications

Work at the Lab

实验室的工作





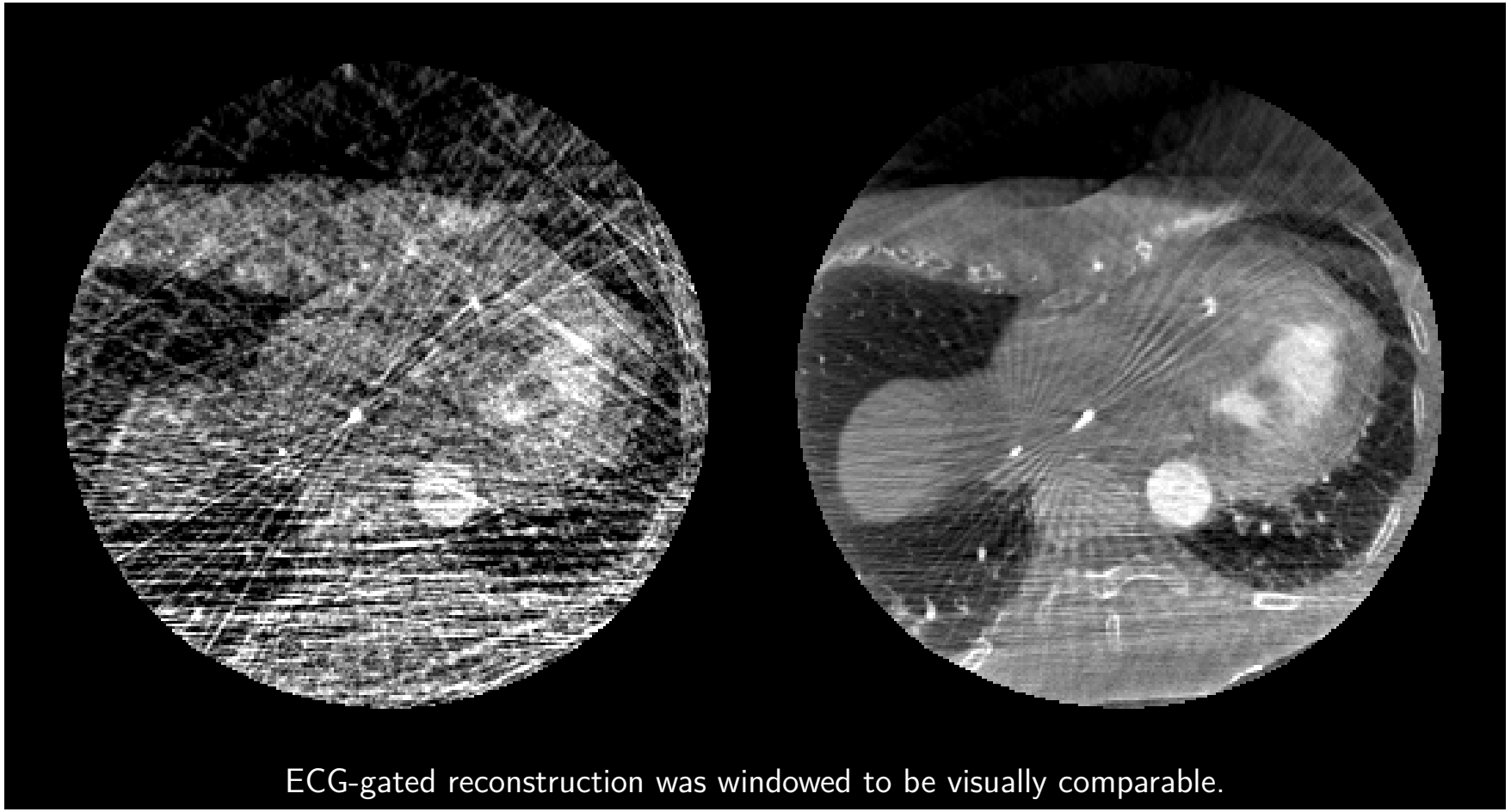






Motion Compensation

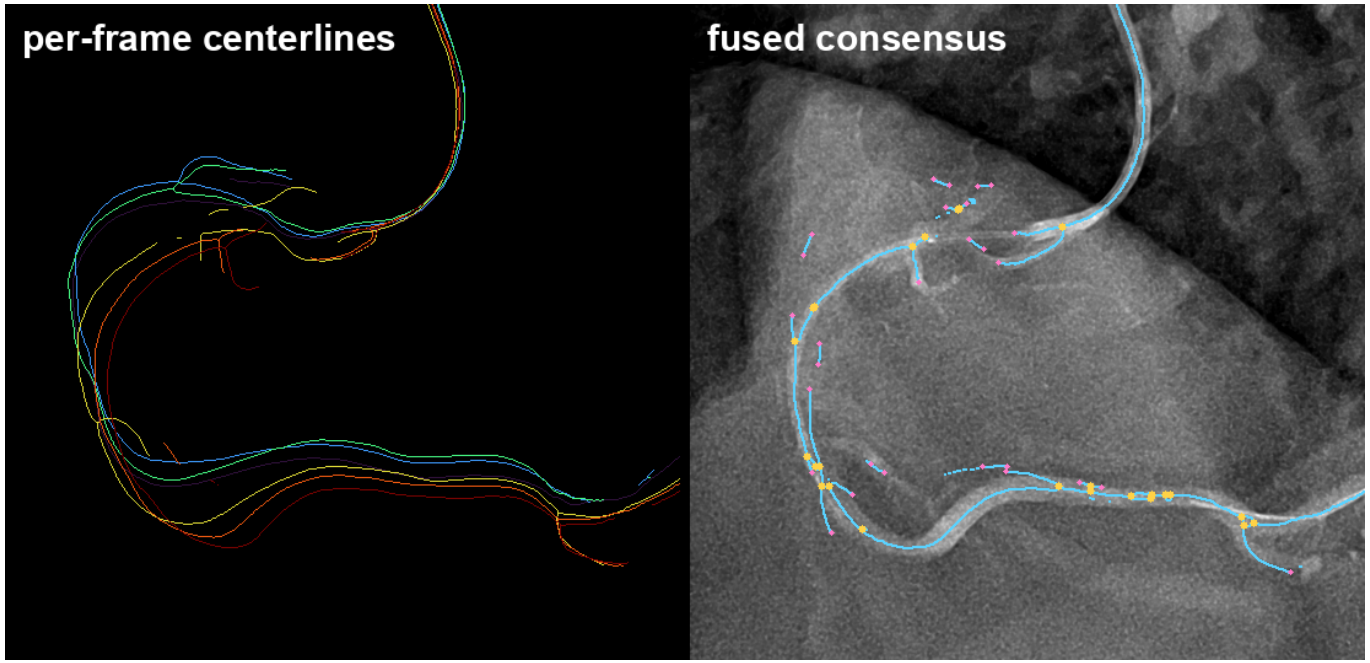
运动补偿



ECG-gated reconstruction was windowed to be visually comparable.

VesselROVER

Coronary centerlines in angiography 冠状动脉中心线提取



Left: per-frame centerlines of one run, spread by cardiac motion. Right: fused consensus centerline and branch points on a clinical frame.

- No single frame shows the complete coronary tree.
- Fuse several segmenters (STAPLE) *and* many frames (ROVER, from speech recognition).
- 13 patients, ∼300 clinical runs: centerline tree **1.22**× more complete, **6.7**× more distal branches.
- Open source: `github.com/akmaier/VesselROVER`

Joint work with Dr. Wenzheng Han, Shanghai Chest Hospital

A. K. Maier, S. Bayer, W. Han, *Combining rater fusion and time fusion for coronary centerline extraction from single-plane angiography*, SPIE Medical Imaging 2027 (accepted).

Outline

提纲

- What is deep learning?
- What is agentic AI?
- What does it mean for science?

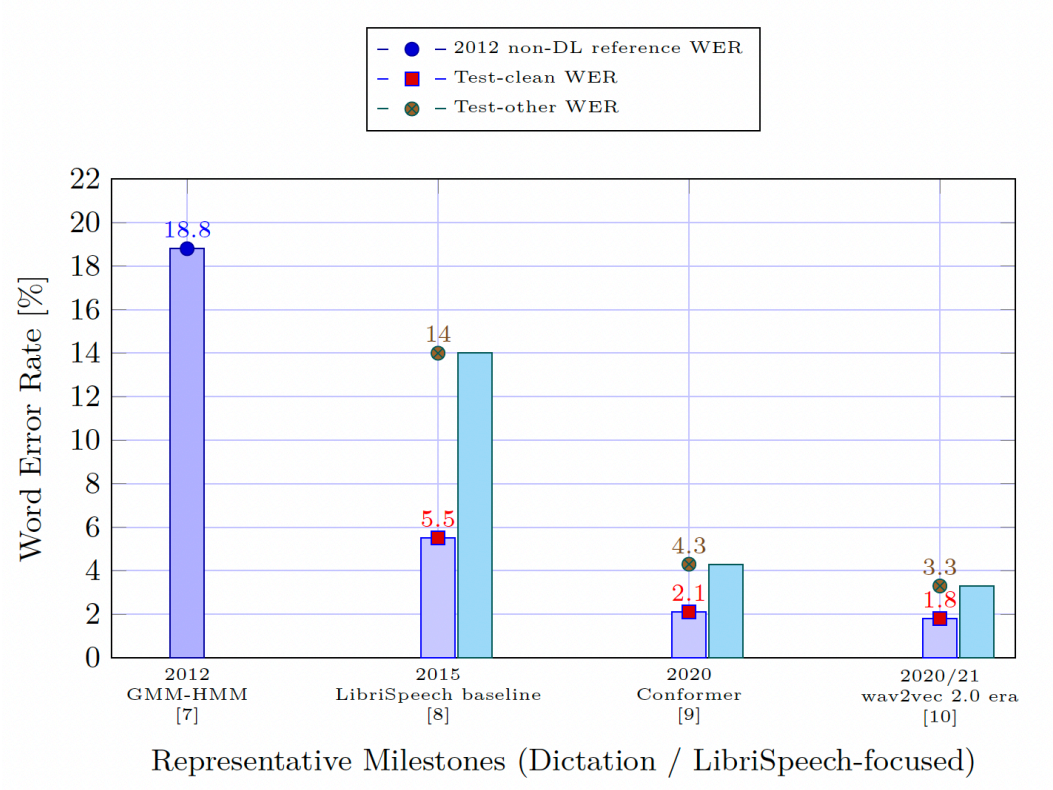
What is deep learning?

什么是深度学习?

Deep learning

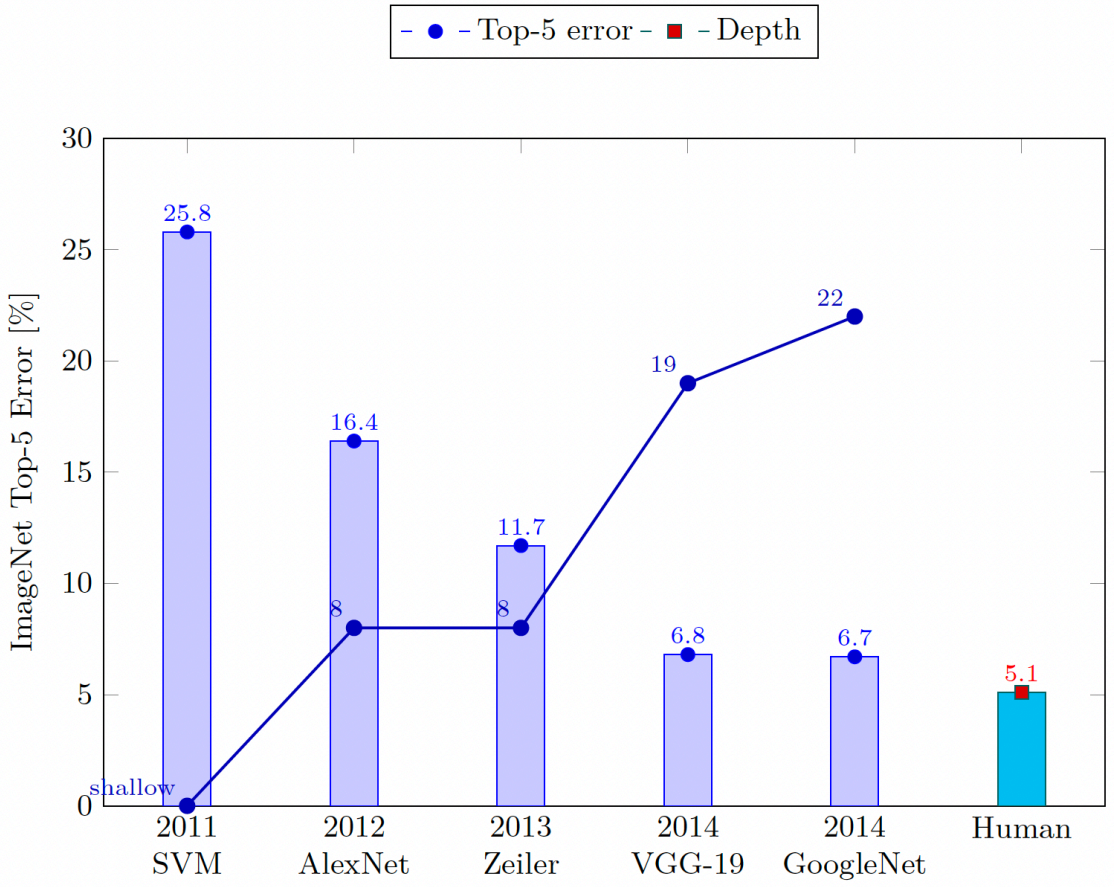
Speech 深度学习

- Revolutionising pattern recognition and AI for more than 10 years.



Deep learning

Image classification 深度学习



Deep learning

Learning to play 深度学习



[Mnih 2013]

- Deep Q-Networks learn Atari games from raw pixels alone.
- One agent, many games – super-human on several titles.

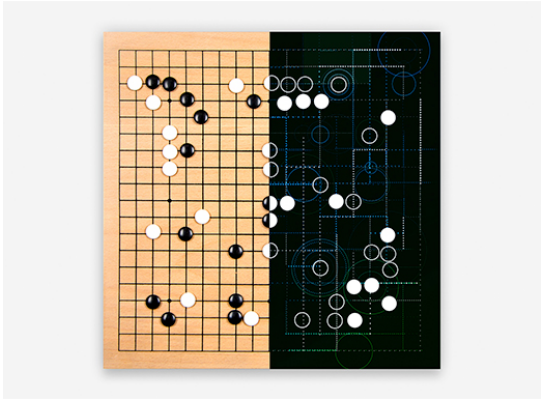
Deep learning

Learning to play 深度学习



[Mnih 2013]

- Deep Q-Networks learn Atari games from raw pixels alone.
- One agent, many games – super-human on several titles.



AlphaGo (DeepMind, 2016)

- Self-play + Monte-Carlo tree search.
- Defeats world champion Lee Sedol 4–1.

Generative AI

Source image 生成式人工智能



Generative AI

Image description 生成式人工智能



"An athletically built man with a serious expression stands in a well-lit room with a wooden ceiling. He is wearing a blue t-shirt with the phrase 'SAVE THE CHUBBY UNICORNS' and an image of a rhinoceros. The man is taking a selfie with a modern smartphone, capturing his reflection in a mirror. In the background, one can see gym equipment including dumbbells and a weight rack. There is also a hanging chandelier with a unique design, illuminating the room."

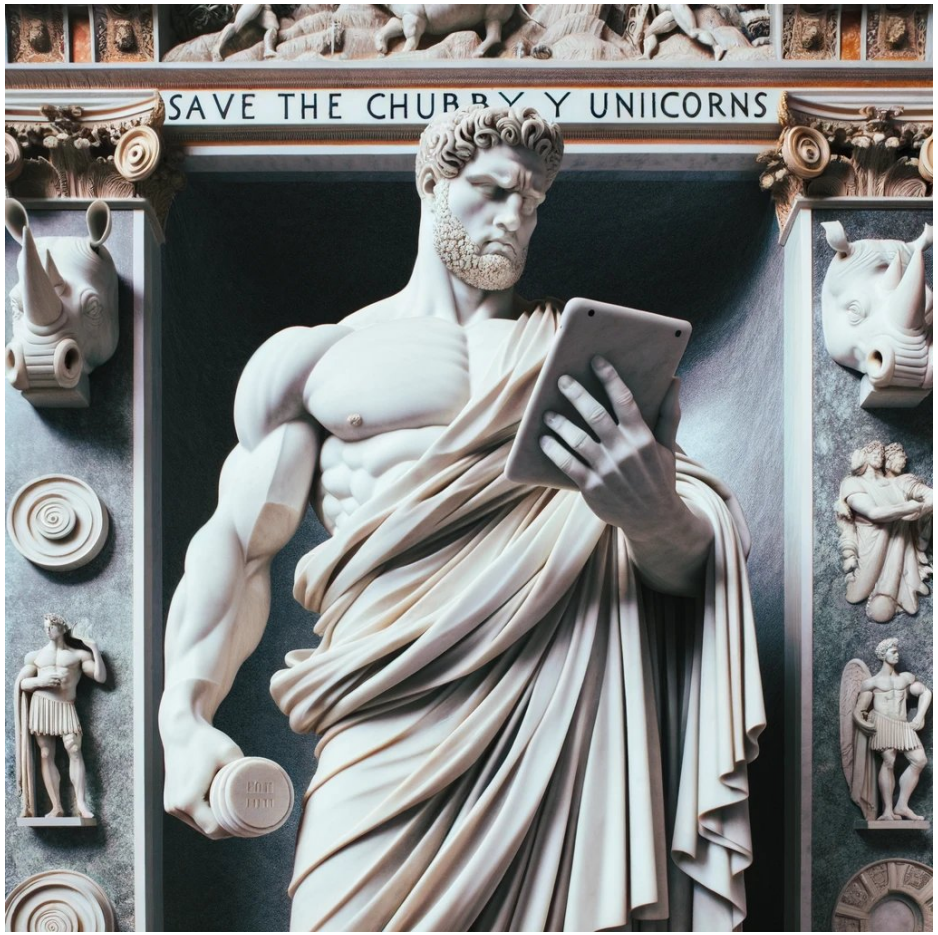
Generative AI

Generated image 生成式人工智能



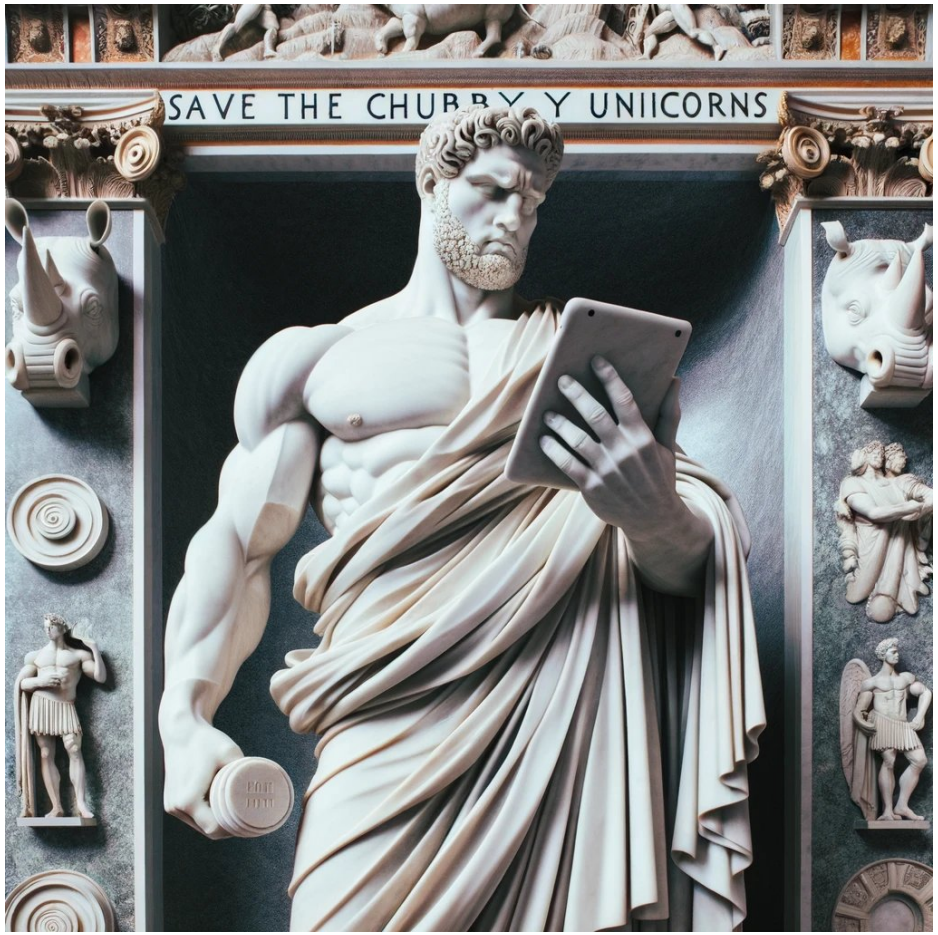
Generative AI

Style variations 生成式人工智能



Generative AI

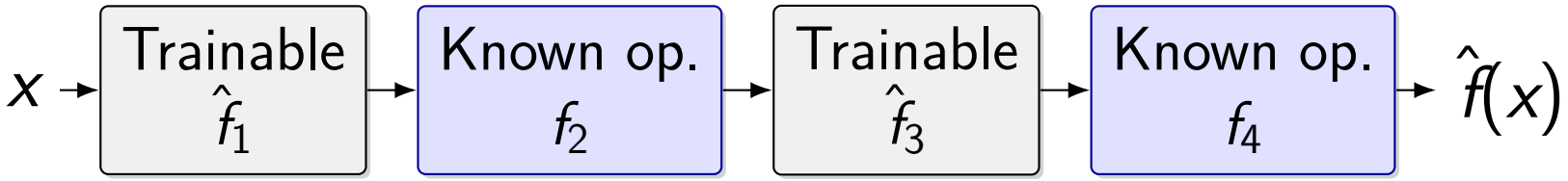
Style variations 生成式人工智能





Deep learning in medicine

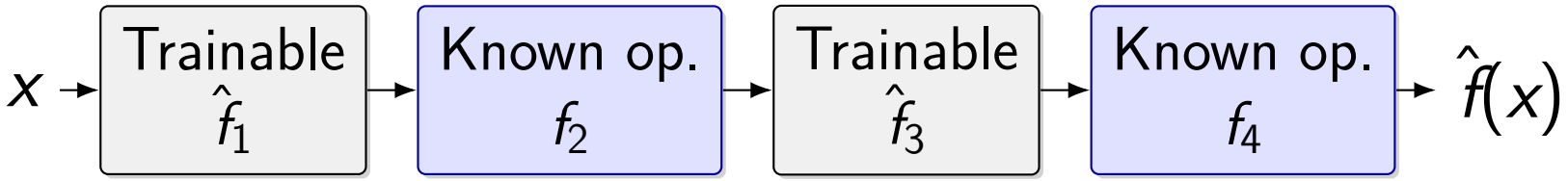
Known operators 医学中的深度学习



Mix *trainable* layers with deterministic operators (Fourier transform, gradient descent, projection, ...).

Deep learning in medicine

Known operators 医学中的深度学习



Mix *trainable* layers with deterministic operators (Fourier transform, gradient descent, projection, ...).

Theorem – known operators reduce the error bound

For $f = f_n \circ \cdots \circ f_1$ approximated by $\hat{f} = \hat{f}_n \circ \cdots \circ \hat{f}_1$ with per-layer errors ε_i and Lipschitz constants $\hat{L}_j$,

$$\varepsilon(\hat{f}) \leq \sum_{i=1}^{n} \left(\prod_{j=i+1}^{n} \hat{L}_j \right) \varepsilon_i .$$

Replacing $\hat{f}_k$ with the *known* f_k forces $\varepsilon_k = 0$ and strictly tightens this bound.

Maier et al., *Nature Machine Intelligence*, 2019.

Deep learning in medicine

Recognition 医学中的深度学习





Deutscher Zukunftspreis – low-field MRI made possible by deep learning.

What is agentic AI?

什么是智能体 AI?

What is agentic AI?

A simple question 什么是智能体 AI?

“How do you drive a car?”

Thought for a few seconds...

- A classic prompt to a large language model.
- After a short “thinking pause” the answer arrives as long-form running text.
- No tools, no actions – the model just *describes* how to drive.

What is agentic AI?

Long text answer of an LLM 什么是智能体 AI?

To drive a car, follow these steps:

- **Get in and adjust:** fasten your **seat belt**, set the **seat** and **mirrors** so you can see the road clearly.
- **Start the engine:** press the **brake pedal**, turn the key or push the start button. Make sure the gear selector is in **Park (P)** or **Neutral (N)**.
- **Engage a gear:** for an automatic, move the lever to **Drive (D)**; for a manual, press the **clutch** and select **1st gear**.
- **Release the parking brake** and slowly lift your foot off the brake. Gently press the **accelerator** to move forward.
- **Steer** smoothly with both hands and **check mirrors** regularly. Keep a safe distance from the vehicle in front.
- **To stop:** ease off the accelerator, press the brake, and bring the car to a complete stop. Shift to **Park** and apply the parking brake before turning off the engine.

Always observe traffic rules and adapt your driving to road and weather.

$\Rightarrow$ **Lots of words, but no *action*.**

What is agentic AI?

From language model to agent 什么是智能体 AI?



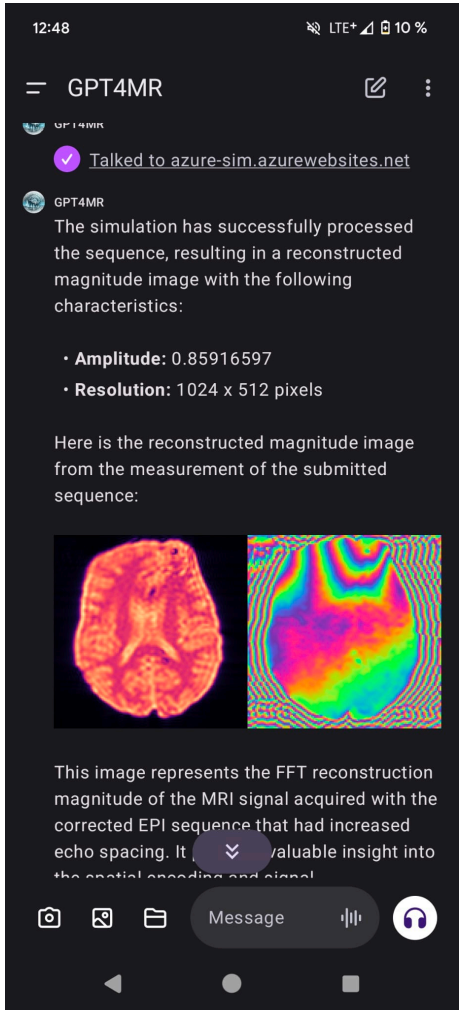
Moritz Zaiss

What is agentic AI?

From language model to agent 什么是智能体 AI?



Moritz Zaiss



Zaiss et al., *GPT4MR*, 2024.

What is agentic AI?

Task for the models 什么是智能体 AI?

"Code a spin echo EPI (64×64, fov = (0.2, 0.2, 0.08) m, TE = 150 ms)!"

- **Spin-echo EPI** – an advanced MRI sequence used for diffusion and functional imaging.
- Three models receive the *same* natural-language instruction and must produce runnable Python code.
- Contestants:

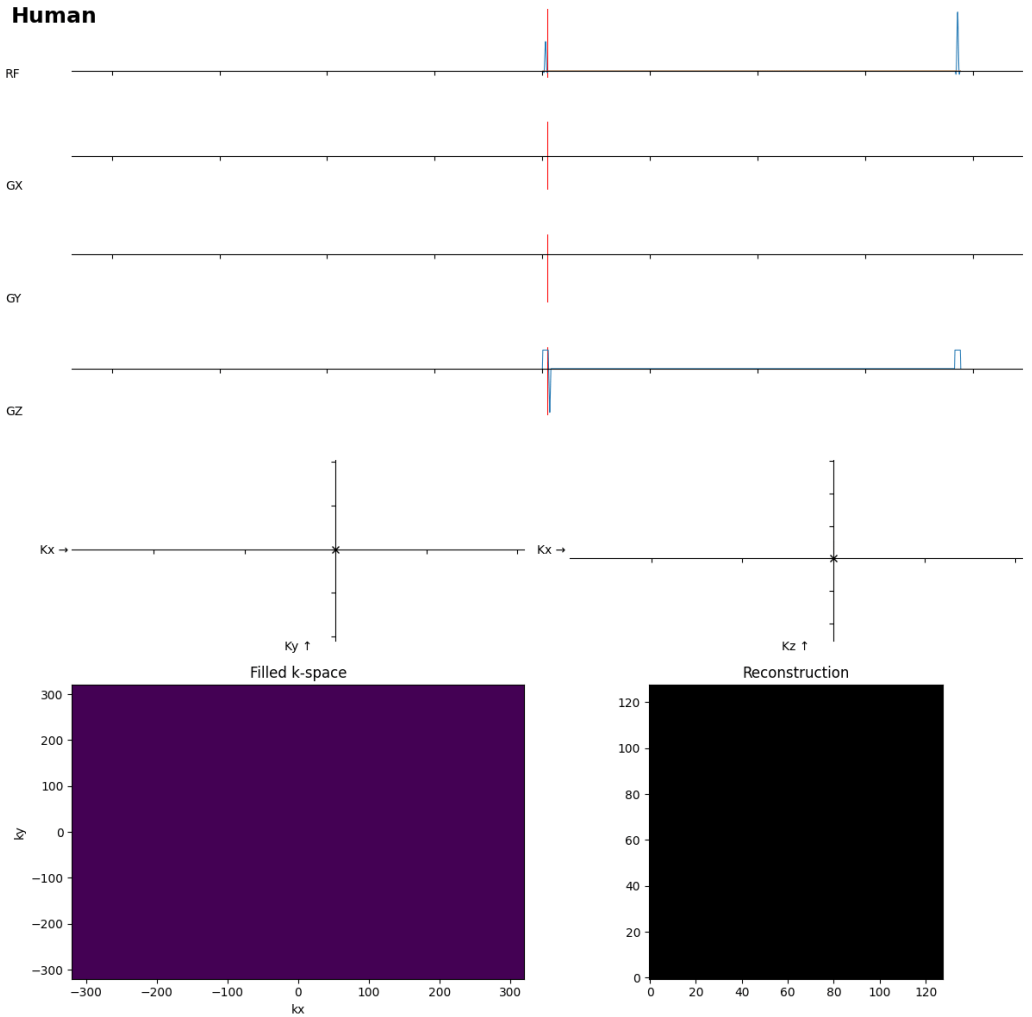
Human
reference

GPT-4o
OpenAI

GROK-3-think
xAI

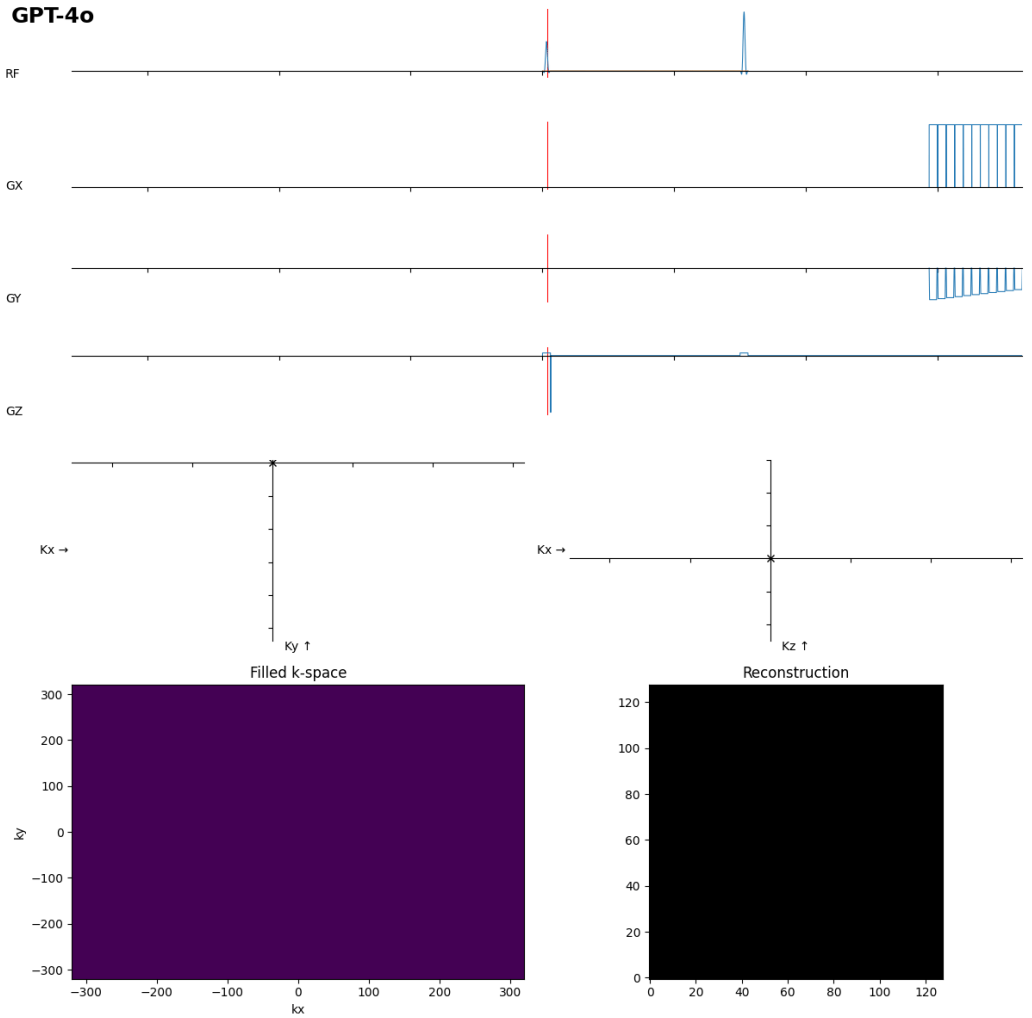
What is agentic AI?

Human baseline 什么是智能体 AI?



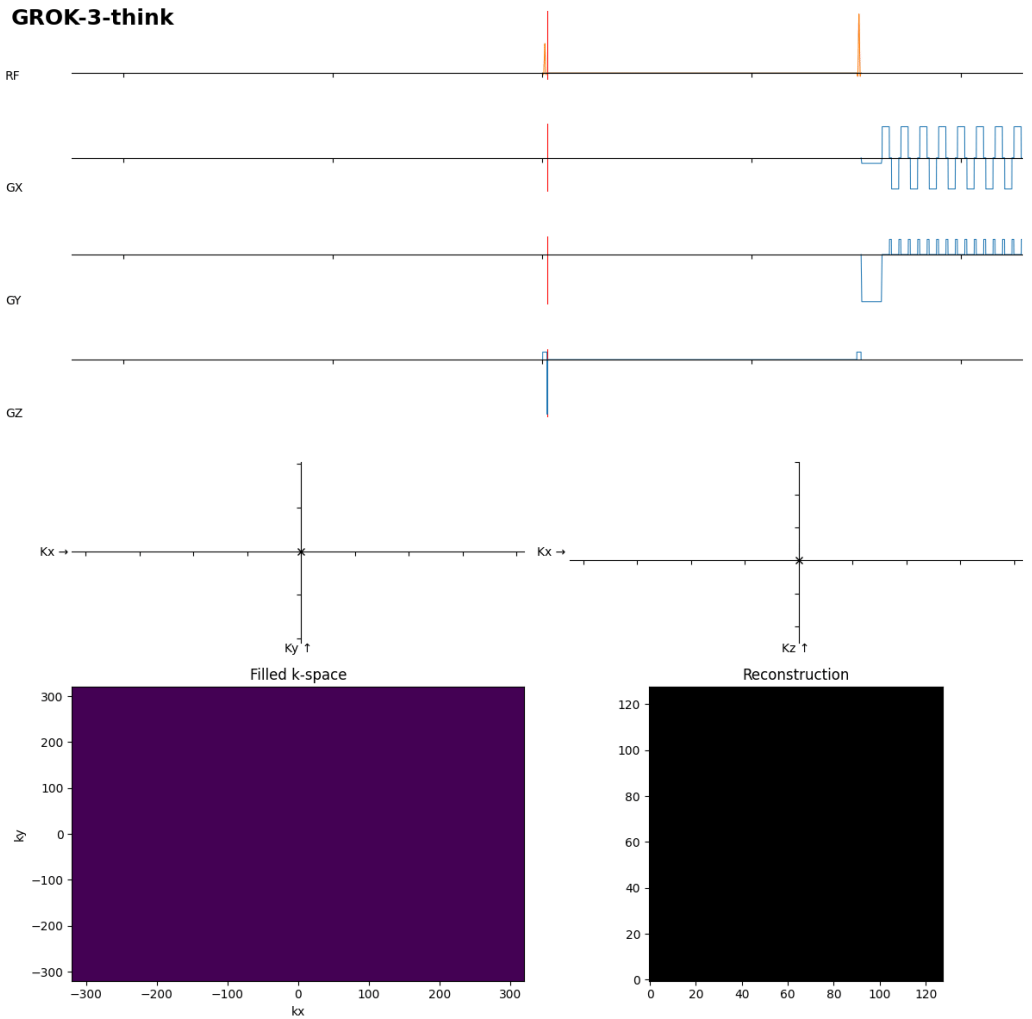
What is agentic AI?

GPT-4o attempt 什么是智能体 AI?



What is agentic AI?

GROK-3-think attempt 什么是智能体 AI?

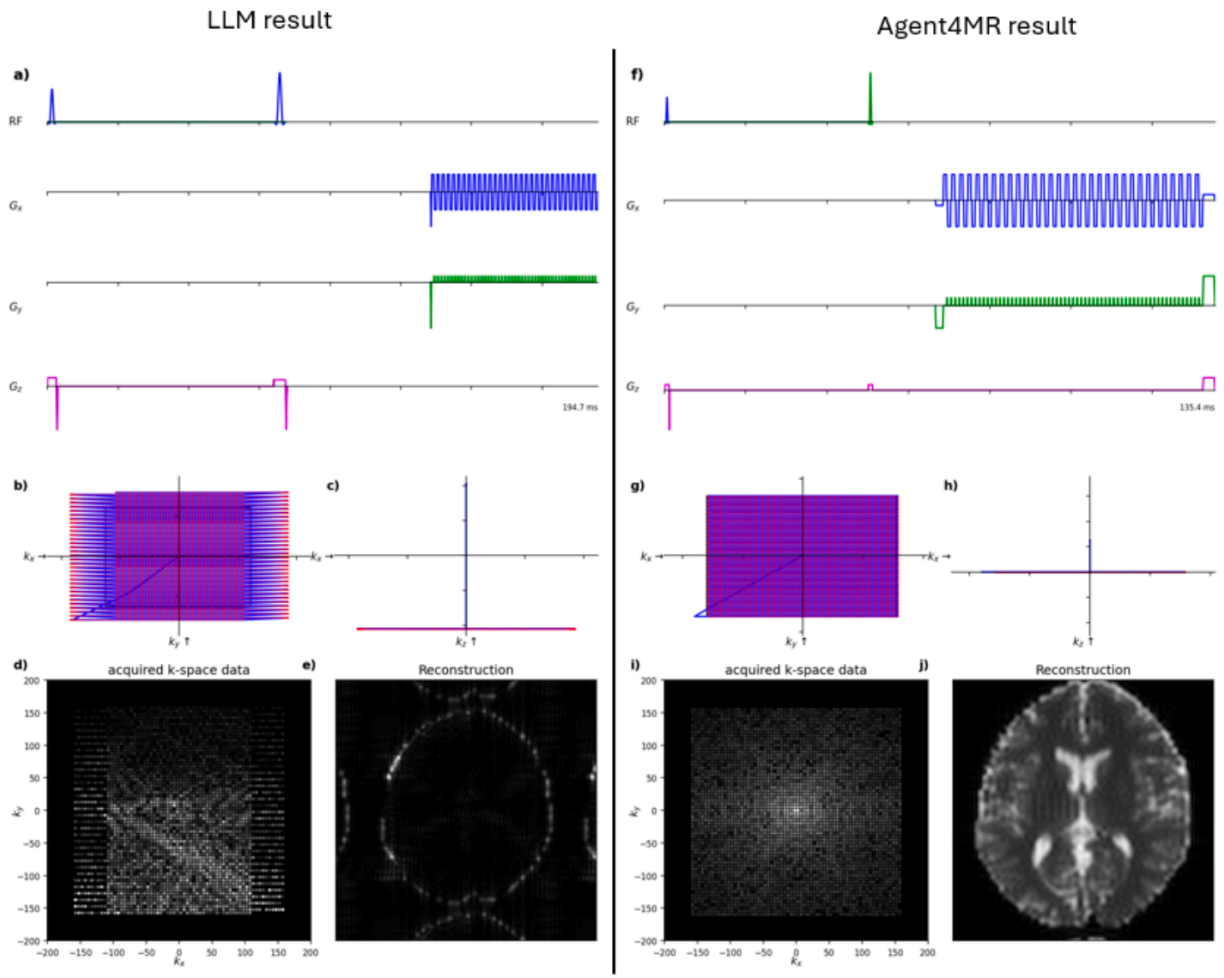


How about real agentic AI?

真正的智能体 AI 又如何?

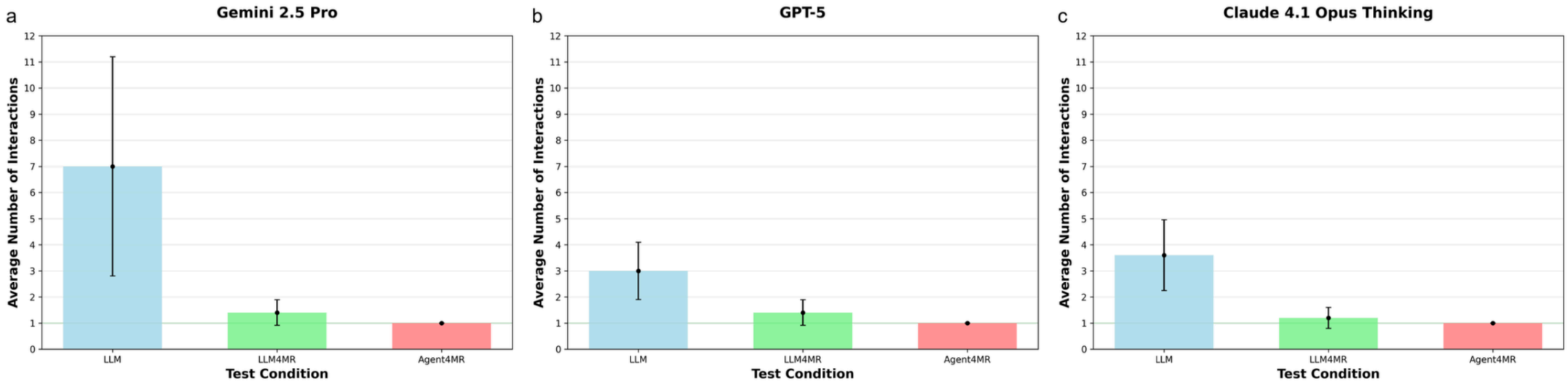
What is agentic AI?

LLM vs. Agent4MR 什么是智能体 AI?



What is agentic AI?

Interactions per model 什么是智能体 AI?



What is agentic AI?

Cost comparison 什么是智能体 AI?

Table 1: User interactions N, time, token usage, and estimated cost until a satisfactory sequence. Human baseline: recently trained EPI-naive student, 2.5 h, 5 interactions.

Configuration Agent4MR	N	Time [min]	Tokens	Cost [€] [≈]
Gemini 2.5 Pro	1	4.8	87k	0.12
GPT-5	1	15	520k	0.48
Claude 4.1 Opus	1	10.2	1162k	10.00
Human	5	150	—	37.50

What is agentic AI?

Iterative progress 什么是智能体 AI?

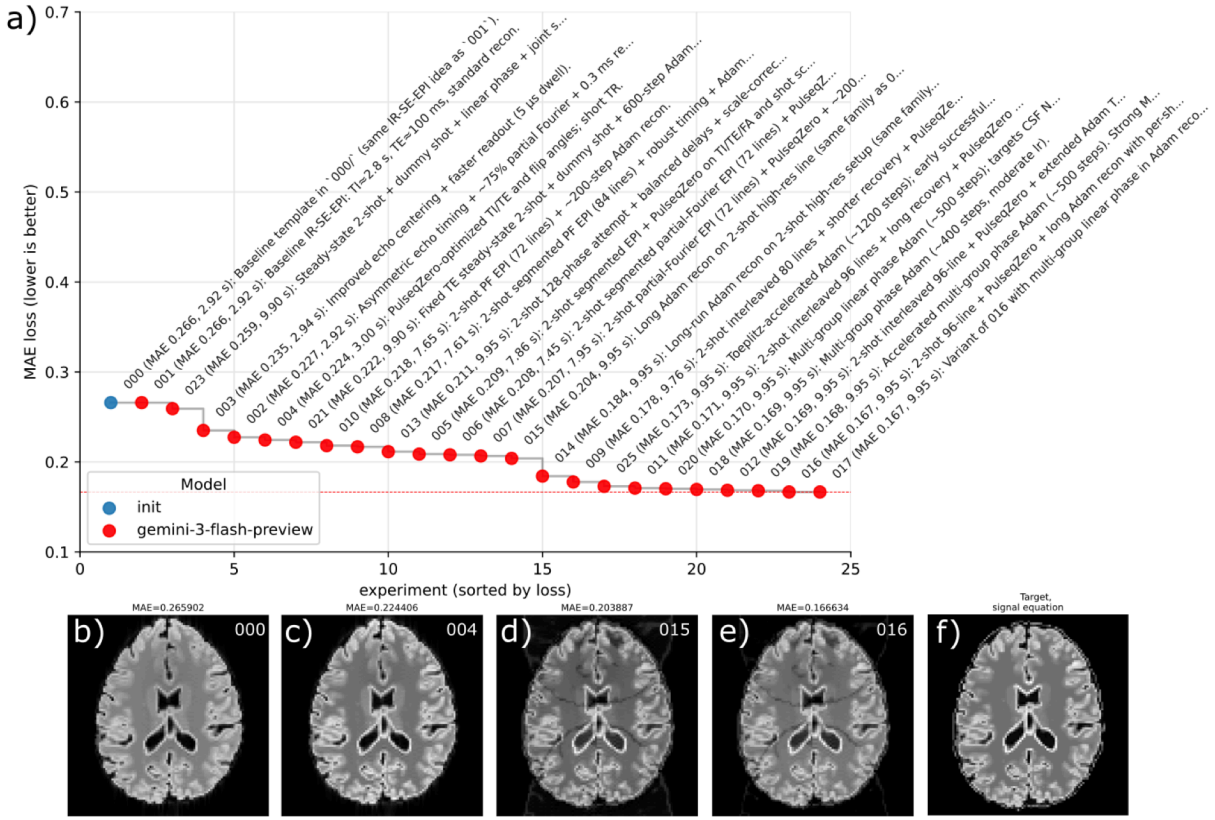


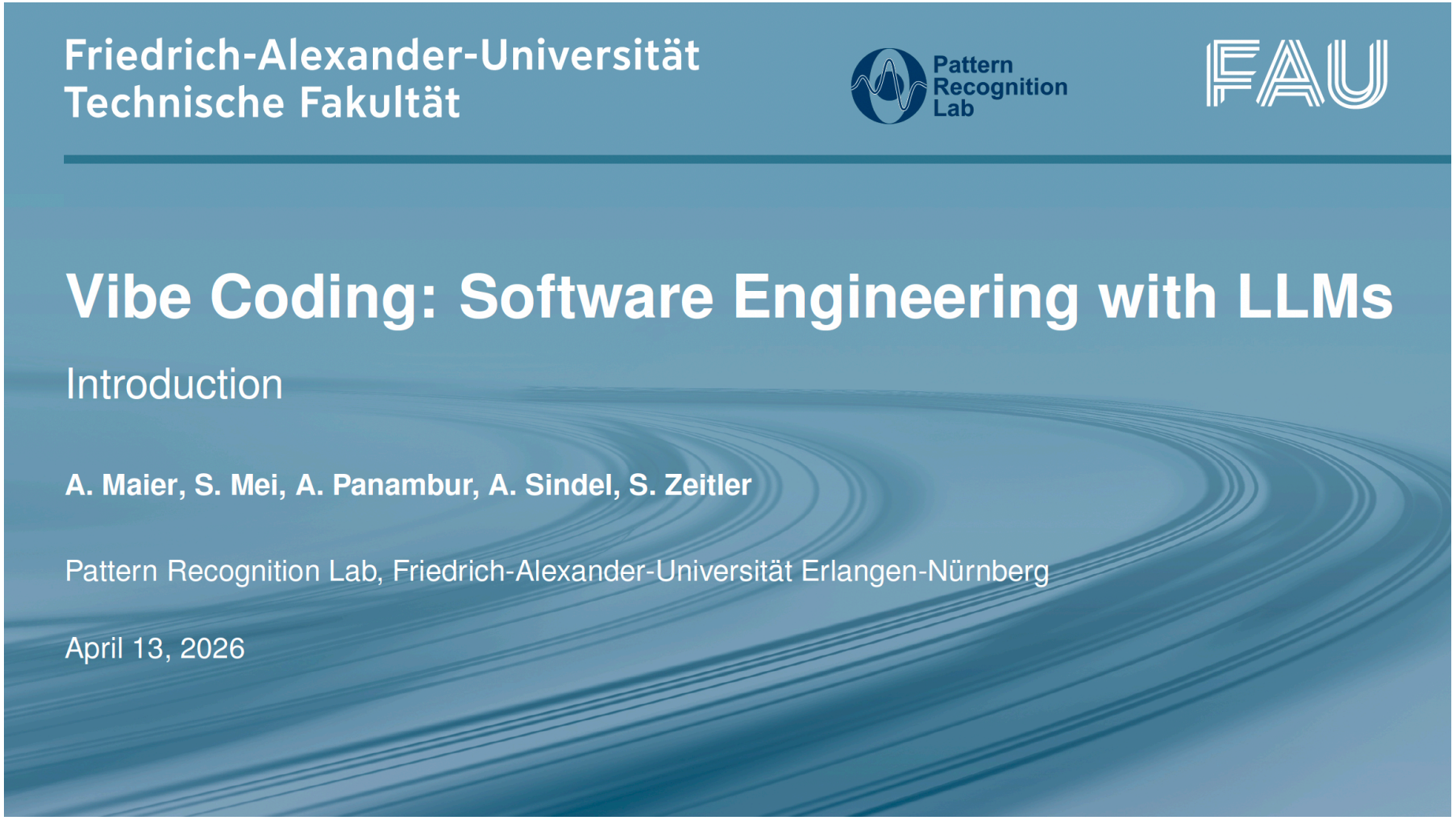
Figure 5: IR-SE-EPI MAE leaderboard progress. Autonomous agents iteratively improved the IR-SE-EPI pipeline (sequence parameters, reconstruction, post-processing). Experiments are ranked by MAE loss (worst to best). The staircase shows improvement from baseline (∼0.2659) to best (∼0.1666) through multi-window filtering, multi-shot introduction, phase correction, and differentiable TI/TE optimization.

Table 2: Normalized Token Usage and API Costs in EUR (n=25 events per model)

Gemini Model	Total Tokens [Mio]	Tokens per Exp. [Mio]	Cost [€]
2.5 Flash	25.456	1.018	2.43
3 Flash	136.340	5.453	12.85
3.1 Pro	72.039	2.881	59.72
Total	**233.836**	**9.353**	**75.00**

What is agentic AI?

Vibe coding 什么是智能体 AI?



What is agentic AI?

Advanced AI-assisted techniques 什么是智能体 AI?

Advanced AI-Assisted Techniques

Claude Skills [3]:

- Shareable workflow automation (YAML + Markdown)
- Team-wide best practices as code
- Deployment, testing, code review
- Available at `github.com/anthropics/skills`

Model Context Protocol (MCP):

- Standardized tool integration
- Connect AI to databases, APIs, files
- Growing ecosystem of plugins

Other Advanced Patterns:

- **Agentic workflows**: Multi-step autonomous tasks
- **Tool-augmented coding**: AI uses external tools
- **Vision support**: Screenshot-driven debugging

Key Benefit

These techniques enable AI to perform complex, multi-step workflows beyond simple code generation.

to a programmer interface for an agent.

Pattern Recognition Lab A. Maier et al. Vibe Coding: Software Engineering with LLMs – Introduction December 2, 2025 11/27

What is agentic AI?

Markdown-driven quality gates 什么是智能体 AI?

Geek Box 13.3: Markdown-driven quality gates: when agents run the production checklist

A key insight from agentic software engineering is that quality criteria, which used to require human inspection, can now be expressed as structured markdown instructions that an AI agent executes autonomously. The agent reads the checklist, runs the specified checks, collects all findings, and presents them for human review—combining the thoroughness of automated scanning with the judgment-level assessments previously exclusive to human reviewers.

Example: a production checklist for book chapters. This book uses a markdown-based production checklist that an AI coding agent runs before each chapter is finalized. A condensed excerpt:

```
# Production Checklist for Chapter Review
The agent should: 1. Detect all problems first.
2. Present a PASS/FAIL table. 3. Ask the user
which to fix. 4. Apply only selected fixes.

## Checks to Run
### Cross-References
- All figures/tables/geek boxes referenced in text.
### Accessibility
- All figures have descriptive alt-text.
### Layout
- No "Float too large" or overfull hbox warnings.
### Acronyms
- No duplicate acronym expansions.
```

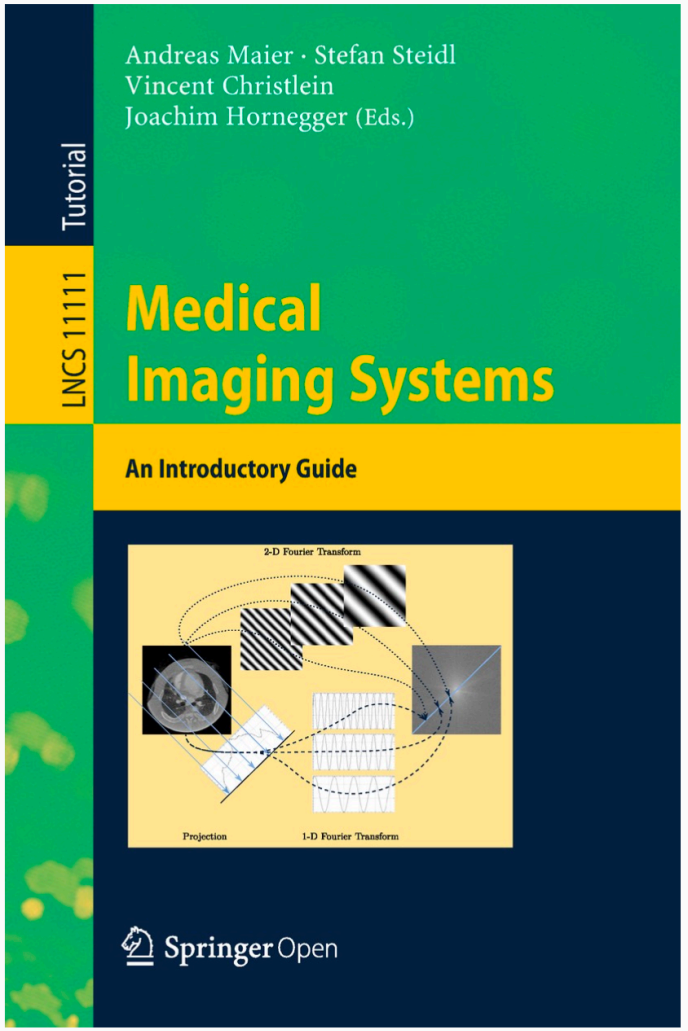
Several of these checks—alt-text quality, duplicate acronym detection, caption descriptiveness—require understanding the *meaning* of text, not just its syntax. An AI agent can evaluate whether alt-text actually describes what a figure shows or whether a manually spelled-out acronym conflicts with a `\gls{}` expansion nearby. These are assessments no simple script can perform reliably.

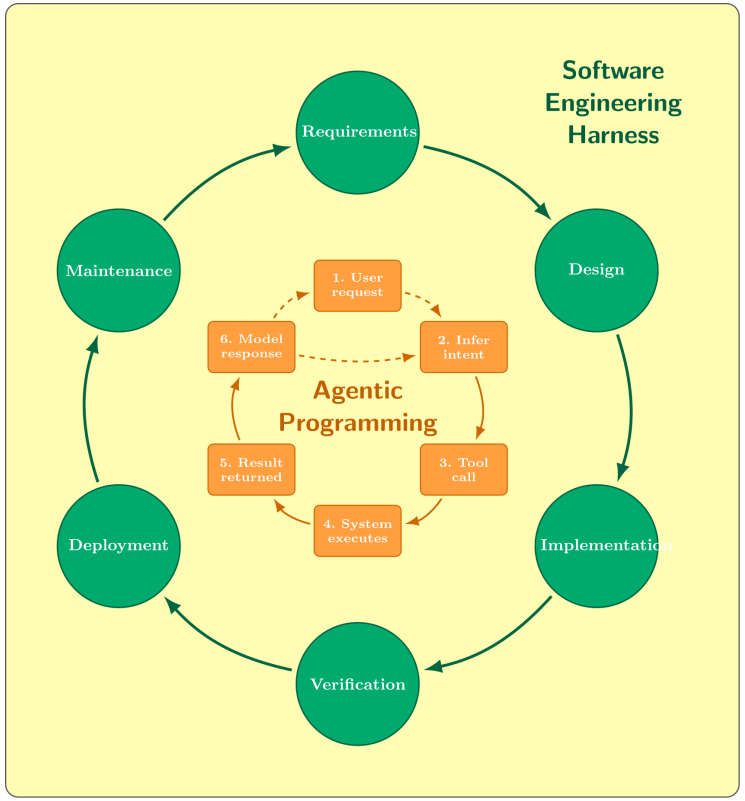
The detect-then-ask workflow. The checklist specifies a four-step process: detect all issues, present a PASS/FAIL summary, ask the user which fixes to apply, and then execute only approved changes. This separation of detection from correction mirrors the role-separation principle from Chapter 12: the agent that finds problems should not unilaterally fix them. The human decides which findings are genuine issues and which are acceptable.

This pattern generalizes beyond book production. Any domain with explicit quality criteria—code review checklists, regulatory compliance, accessibility audits, documentation reviews—can be encoded as markdown and executed by an agent. The quality gate becomes a reusable, version-controlled artifact rather than tacit knowledge in a reviewer's head.

What is agentic AI?

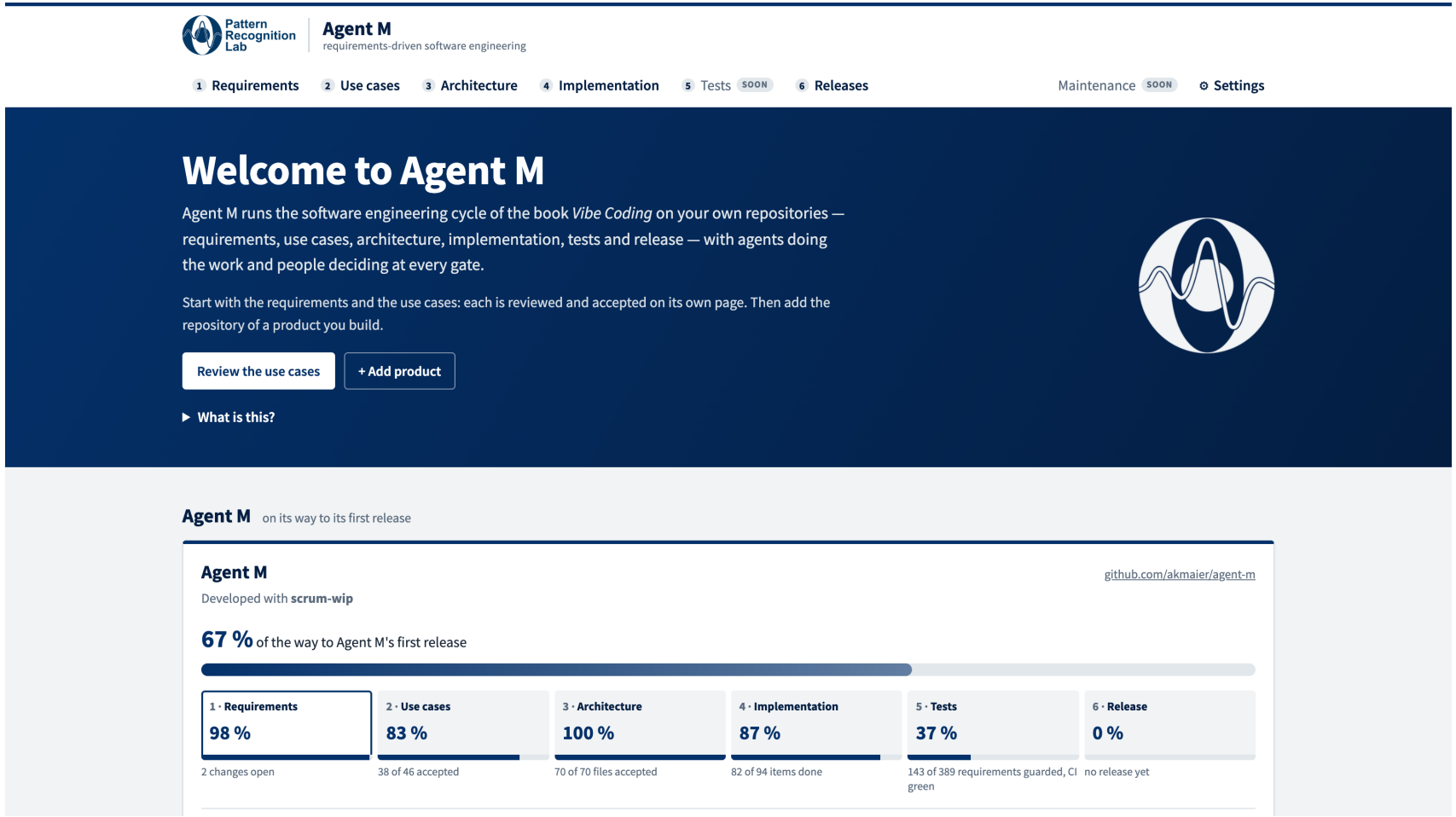
Textbook 什么是智能体 AI?





What is agentic AI?

Agent M – companion tool to *Vibe Coding* 什么是智能体 AI?



akmaier.github.io/agent-m

What does it mean for science?

这对科学意味着什么?

Almost everything is easy now...

China and AI 如今几乎一切都变得容易……



Almost everything is easy now...

Conference shares 如今几乎一切都变得容易……

Conference	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
CVPR	~30%	32%	35%	35%	39%	39.2%	~40%	~38%	~39%	~40%
ICCV/ECCV	~30%	30%	32%	35%	~37%	~40%	~40%	~38%	~40%	~40%
ICML	~10%	12%	15%	18%	~20%	~22%	~25%	~28%	~30%	~30%
NeurIPS	8%	10%	12%	15%	*~15%*	~18%	~20%	~25%	~28%	~30%
AAAI	15%	18%	22%	28%	~35%	50%	~50%	~45%	~48%	~50%

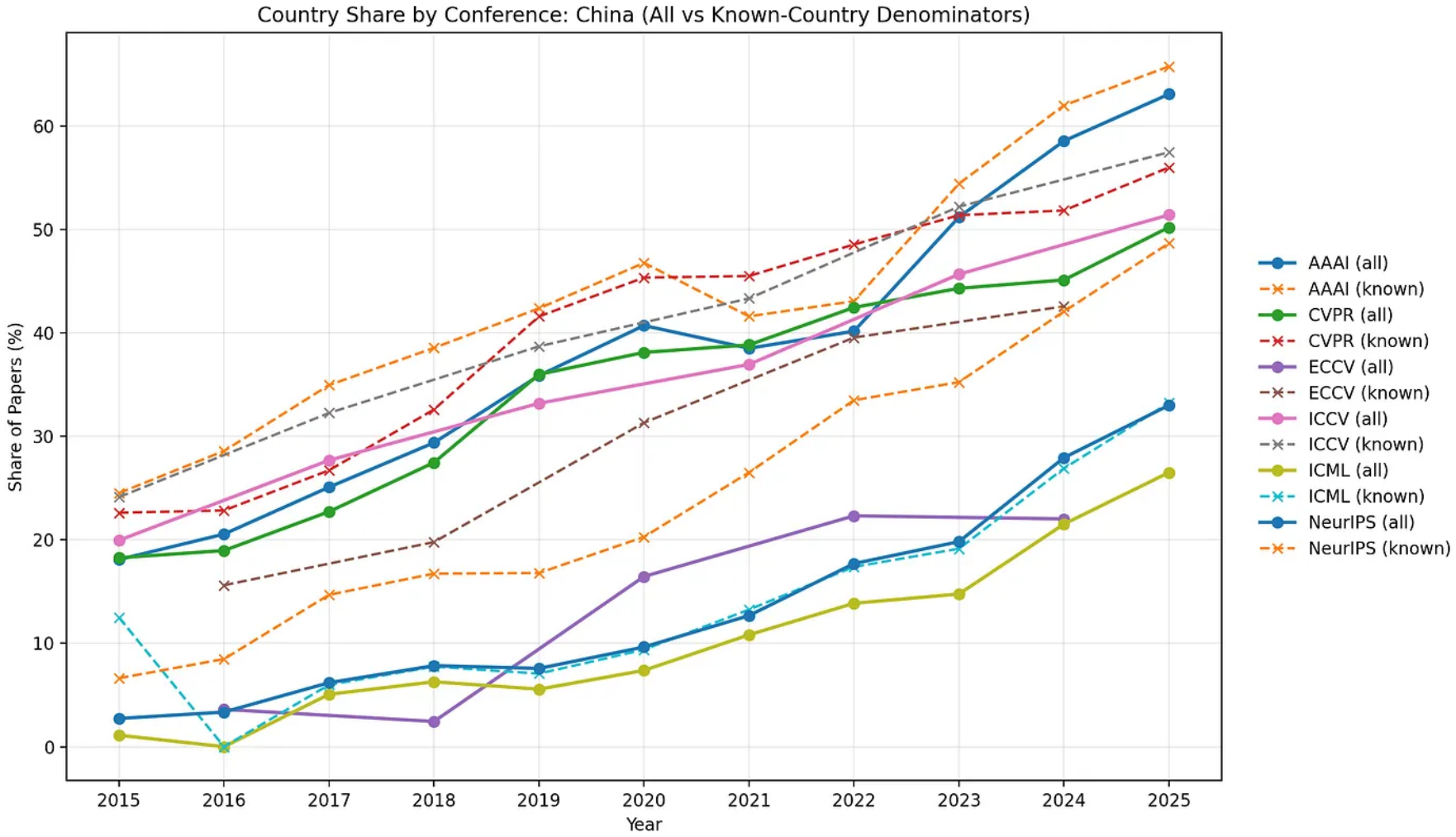
Almost everything is easy now...

ConferenceStats workflow 如今几乎一切都变得容易……



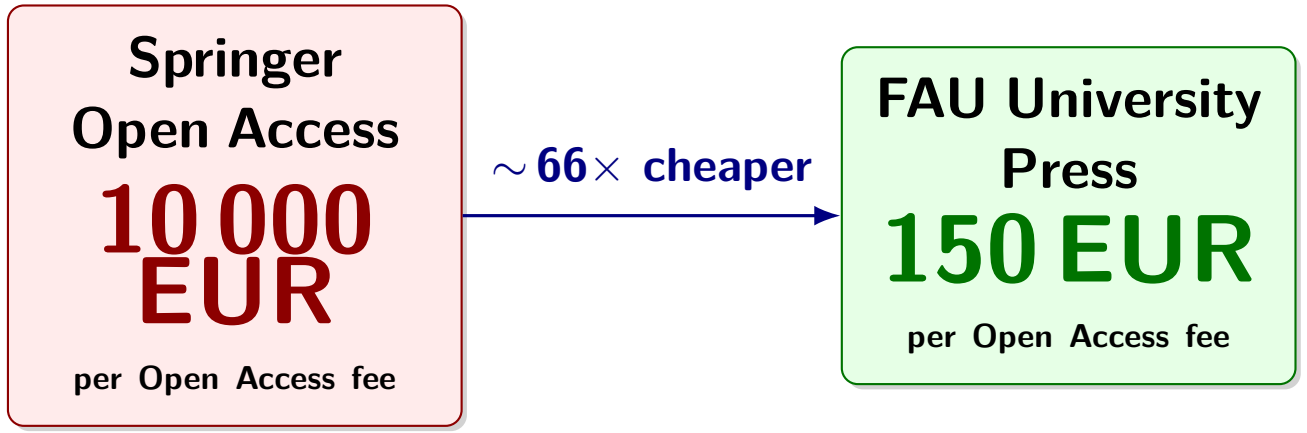
Almost everything is easy now...

Conference contributions from China 如今几乎一切都变得容易……



Publishing, Disrupted

Open Access for the price of a pizza party 出版业的变革



- **106** submissions → **93** accepted → one Open Access volume.
- Idea to printed book: **three months**, including **peer review**.

Agent4CT

All deep-learning CT methods, in one weekend 智能体复现 CT 方法

- Continuously-running LLM agent that edits a CT reconstruction codebase, runs short GPU experiments, and keeps changes only when the metrics agree. Five agents run in parallel across five CT challenges and pool findings through a shared scratch pad.

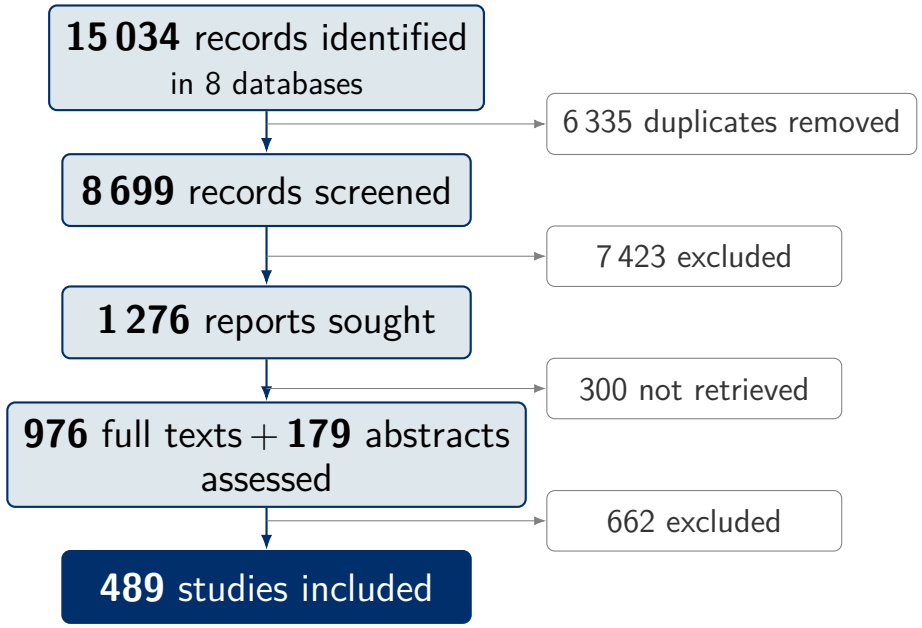
Re-implemented solvers (`pentathlon/demo_dl_reference`):

- **Analytical:** PyroNN FBP (Syben 2019).
- **Classical iterative:** TV (iterative, supervised, hyper-search); Wu *et al.* 2015 (+ trainable port).
- **Variational / unrolled:** Hammernik 2017 variational network (+ CT port); Learned Primal-Dual (Adler & Öktem); ITNet v1–v3.
- **Self-supervised denoising:** dual-domain U-Net and trainable bilateral filters (Wagner 2023 / 2022, with supervised variants).
- **Per-scene neural fields:** NAF (neural attenuation field); R2Gaussian.
- **Foundation models:** RAM (Terris 2025).
- **Diffusion priors:** DDPM; diffusion-prior sampling; diffusion-recon with data consistency.
- **Transformer:** U-Swin.

`github.com/akmaier/Agent4CT`

A Living Systematic Review

Robotic active matter – PRISMA 2020 活体系统综述



- First review cycle in **seven days** (11–17 Oct 2026).
- Search in OpenAlex, NASA ADS, PubMed, IEEE Xplore, arXiv, Crossref, HAL and bioRxiv; protocol registered on OSF.
- Screening and extraction by a **three-LLM panel** on NHR@FAU – every vote is public.
- Author-validated: sensitivity **1.00** on a 200-record sample.
- **Living**: anyone can propose a missing paper by pull request.

`akmaier.github.io/ActiveMatterLivingReview`

Thanks to Siming Bayer for the idea!

Speed of Agentic AI

Wall-clock from idea to working artefact　智能体 AI 的速度

3 h Reproduce a paper + code, find new insights, draft a follow-up paper.
arxiv.org/abs/2605.02620

1 d Reproduce a paper *without* its code or data.
arxiv.org/abs/2605.12772

3 d Scrape 200 study programs + ship a RAG chatbot.
github.com/akmaier/IfCampoKnew

1 w Reimplement an entire ATARI VCS emulator – bit-exact, 16 k lines of code.
github.com/akmaier/UnderstandingVCS

Science at Agent Speed

Opportunities and risks 智能体速度下的科学

Opportunities

- Months of work in hours.
- Reproduce papers without code.
- Open Access for 150 EUR.
- Coding for every discipline.

Risks

- Plausible is not correct.
- Hallucinated references.
- AI-text detectors beaten in 3 h.
- Sponsored bias in 12 LLMs.

Maier et al., arXiv:2605.02620 (style detector) ▪ Maier et al., arXiv:2605.12772 (sponsored recommendations)

What This Means for Your PhD

Five rules for working with agents 这对你的博士研究意味着什么

1. Be the verifier, not the typist.
2. Reproduce before you build.
3. Keep a git-backed lab journal.
4. Write quality gates, not just prompts.
5. Disclose AI use – own the final word.

References

参考文献

- V. Mnih et al., *Playing Atari with Deep Reinforcement Learning*, arXiv:1312.5602, 2013.
- D. Silver et al., *Mastering the game of Go with deep neural networks and tree search*, *Nature*, 529:484–489, 2016.
- **A. Maier** et al., *Learning with Known Operators Reduces Maximum Error Bounds*, *Nat. Mach. Intell.*, 1:373–380, 2019.
- M. Zaiss, J. R. Rajput, H. N. Dang, V. Golkov, D. Cremers, F. Knoll and **A. Maier**, *Exploring GPT-4 as MR sequence and reconstruction programming assistant: GPT4MR*, BVM Workshop, pp. 94–99, Springer, 2024.
- M. Zaiss, A. Aly, J. Endres, T. Dornstetter, S. Weinmüller and **A. Maier**, *Agentic MR sequence development: leveraging LLMs with MR skills for automatic physics-informed sequence development*, arXiv:2604.13282, 2026 (Agent4MR).
- Deutscher Zukunftspreis (low-field MRI): https://www.deutscher-zukunftspreis.de
- Agent M – companion tool to the book *Vibe Coding*: https://github.com/akmaier/agent-m
- Conference statistics tooling: https://github.com/akmaier/ConferenceStats
- Agent4CT – agentic re-implementation of deep-learning CT methods: https://github.com/akmaier/Agent4CT
- **A. Maier**, M. Zaiss and S. Bayer, *Beating the Style Detector: Three Hours of Agentic Research on the AI-Text Arms Race*, arXiv:2605.02620, 2026.
- **A. Maier**, J. Sopa, G. G. Sahin, P. Perez-Toro and S. Bayer, *Just Ask for a Table: A Thirty-Token User Prompt Defeats Sponsored Recommendations in Twelve LLMs*, arXiv:2605.12772, 2026.
- *IfCampoKnew* – scraper for 200 study programs and an AI RAG chatbot: https://github.com/akmaier/IfCampoKnew
- *UnderstandingVCS* – agent-reimplemented ATARI VCS emulator: https://github.com/akmaier/UnderstandingVCS
- **A. K. Maier**, S. Bayer and W. Han, *Combining rater fusion and time fusion for coronary centerline extraction from single-plane angiography*, SPIE Medical Imaging 2027 (accepted); code: https://github.com/akmaier/VesselROVER
- **A. Maier**, S. Bayer et al., *A Living Systematic Review of Robotic Active Matter*: https://akmaier.github.io/ActiveMatterLivingReview

Thank you! Questions?

谢谢！欢迎提问

Slides on GitHub 幻灯片见 GitHub



github.com/akmaier/AIMed2026Slides/tree/shanghai-sino-euro-2026

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